Master IASD (AI, Systems, Data)

Geometry of Diffusion

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Preface

Throughout my undergraduate and graduate studies in mathematics I have been caught between two main interests: the applied world of machine learning, and the theoretical world of topology and geometry. While this is often viewed as an unusual mix to choose, I have always found it greatly enjoyable.

This resulted in me wanting to answer the question of what is happening to the “shape” of data throughout a machine learning process, and also what the “shape” of data even is.

The value in understanding this is twofold: firstly, to be able to make better models by leveraging our understanding of how they work; and secondly, to be able to understand how to interpret what these models are doing. The dance between naturally arising data and the model trying to fit without knowing how the data is generated is fascinating. My main motivation was to try to understand this.

One big idea is the notion of data lying on some low-dimensional space, which makes both learning and sampling from distributions supported on this lower-dimensional space easier. However, this belief only truly simplifies the system in the small noise regime. All sorts of complex geometry can happen in the higher noise regimes. We do not, as we will see, have a nicely behaved interpolating family of manifolds.

Before doing the IASD M2 programme, I had written a master’s dissertation on diffusion models under the manifold hypothesis, under the supervision of George Deligiannidis. During this, I became deeply interested in dimension estimation of datasets. This, at best vague (and at worst terribly ill-posed) question seemed to me one where an answer would tell us a lot about the natural geometry of data, and how diffusion models are able to make use of this. Almost every paper I found on the interplay between manifolds and diffusion models cited work from Eddie Aamari.

In November 2025 I reached out to Eddie Aamari to further discuss what the geometry of data could look like, and this led to this internship project. The initial goal was to understand the interplay between how geometry affects sampling error and how this could encourage simpler embeddings in a latent diffusion setting. However, given the complexity of work being done, limited time has been spent on investigating auto-encoders so far.

A key challenge of this project has been striking the balance between geometry, where things are well defined, and diffusion of probability densities, where simply by adding noise we violently change the notions of geometry we would like to equip our density with. This trade-off has led to looking at several objects, which while interesting in their own right have not been helpful in bounding sampling error. I do believe that the final part on posterior covariance fields section 5 does provide a new insight on where true complexity lies in the diffusion process.

Abstract

Diffusion based generative models have become ubiquitous over recent years. Their success is often attributed to the geometric structure of natural data; commonly referred to as the manifold hypothesis. In this report we summarise our main work so far on understanding the geometrical structures of densities undergoing diffusion. We examine an anisotropic stability bound, level sets, ridges, and functionals of the posterior covariance. Ultimately, we settle on the spectrum of the posterior covariance being the key object to understanding the geometry throughout the diffusion process. This yields key quantities to understand diffusion's impact on the geometry of densities, and a natural denoising discretisation schedule.

Algebra is the offer made by the devil to the mathematician. The devil says: 'I will give you this powerful machine, it will answer any question you like. All you need to do is give me your soul: give up geometry and you will have this marvellous machine.'

— ***Michael Atiyah***

1 Introduction

1.1 Motivation and Research Question

Generative modelling has undergone rapid development over the last decade. Score-based generative models (SGMs), diffusion models, and more recently flow-matching formulations provide a framework for transforming a simple reference distribution, typically a Gaussian, into a complicated target distribution p_{data}. The central object in these methods is a time-dependent vector field which transports probability mass between the two distributions. In score-based formulations this vector field is determined by the score

$$\nabla_x \log p_t(x),$$

where p_t denotes the distribution obtained by perturbing the data with an appropriate amount of noise. In flow matching, the same transport can be expressed directly through a learned velocity field. For the Gaussian interpolation considered throughout this report, these descriptions are related explicitly through Tweedie's formula; see section 2.

The empirical success of these methods raises a fundamental geometric question:

> *What geometric structure does a diffusion model see as the data are progressively corrupted by noise, and how does this structure influence the dynamics of the generative process?*

This question is motivated in part by the manifold hypothesis. High-dimensional natural data are often assumed to be concentrated near a lower-dimensional structure

$$\mathcal{M} \subset \mathbb{R}^D, \qquad \dim(\mathcal{M}) = d \ll D.$$

In the small-noise regime, this provides a useful geometric description of the data: uncertainty is concentrated predominantly in directions tangent to $\mathcal{M}$, while the normal directions are strongly constrained. It is therefore natural to expect the geometry of $\mathcal{M}$ to influence both the score field and the dynamics of a diffusion model.

The difficulty is that diffusion does not preserve this geometric picture in any straightforward sense. Gaussian perturbation immediately replaces a potentially singular distribution supported on $\mathcal{M}$ by a smooth density on the entire ambient space. As the noise level increases, the notion of a tangent space becomes progressively less canonical, distinct parts of the data distribution can become statistically indistinguishable, and structures which were geometrically distinct at small noise may merge at larger noise. Consequently, the question is not simply how the original manifold $\mathcal{M}$ affects diffusion. Rather, it is how a notion of effective geometry evolves continuously with the noise level.

This report investigates this question through several geometric objects. We first consider the geometry induced by the flow itself, then level sets and density ridges, before ultimately focusing on the posterior covariance of the clean data conditioned on a noisy observation. The posterior covariance provides a continuous, probabilistic description of local geometry which remains meaningful even when a manifold, ridge, or fixed-dimensional tangent space does not.

1.2 Initial Project

The original objective of this research internship was broader than the analysis presented here. We initially aimed to investigate the geometry of both the auto-encoder and the denoising dynamics in latent diffusion models. The motivating idea was that, if the geometry of a target distribution influences quantities such as score-estimation error, discretisation error, or sampling stability, then jointly training an auto-encoder and a diffusion model might favour latent representations with simpler geometric structure.

This would provide a possible theoretical explanation for a phenomenon frequently exploited in latent generative modelling: representing data in a lower-dimensional latent space can substantially simplify the subsequent generative problem. In such a picture, the encoder would not merely compress the data, but could implicitly search for a representation in which the relevant probability distribution is easier to model and transport.

The analysis required to make this argument precise, however, is substantial. In particular, one must first specify what is meant by the geometry of a probability distribution once the distribution has been smoothed by Gaussian noise. Much of the work therefore became focused on the diffusion process itself, independently of any particular auto-encoder architecture.

The central problem became the following. Given a target distribution p_0 and its Gaussian-smoothed distributions p_t, can we construct a geometric quantity which

1. captures the low-dimensional structure of the data at small noise;
2. evolves continuously as the noise level changes;
3. remains defined when the distribution is no longer concentrated near a single manifold;
4. interacts naturally with the score and velocity fields used by diffusion models; and
5. is sufficiently quantitative to say something about sampling or discretisation error?

The objects investigated in this report are successive attempts to satisfy these requirements.

1.3 Existing Geometric Descriptions

There is an extensive literature connecting the geometry of data distributions to statistical estimation and diffusion models. In particular, several recent results exploit low intrinsic dimension, manifold structure, or related regularity assumptions to obtain improved bounds for diffusion-based sampling and estimation (Benton et al., 2024; Potaptchik et al., 2024; Liang et al., 2025).

A natural starting point is therefore to describe the data using the dimension and regularity of an underlying manifold. These quantities are powerful in the small-noise regime, but become problematic once the distribution is treated as an ordinary density on the ambient space.

Dimension. Assigning an intrinsic dimension $d < D$ to a distribution which is absolutely continuous with respect to $\mathbb{R}^D$ is not canonical. Even when a distribution is concentrated near a manifold, several difficulties arise:

- the effective dimension may vary spatially;
- different regions of the support may have different intrinsic dimensions;
- intersections of geometric structures need not possess a unique local dimension; and
- after Gaussian smoothing, the distribution is full-dimensional for every $t > 0$, so its intrinsic dimension is formally D even when it remains strongly concentrated around a lower-dimensional structure.

Thus, the integer dimension of an underlying manifold does not describe how many directions of uncertainty remain at a particular noise scale.

Reach and curvature. Geometric regularity can also be described using quantities such as reach or curvature (Federer, 1959). These provide strong control over the local geometry of an embedded manifold, but are fundamentally worst-case quantities. They can become degenerate at intersections, corners, or other singular configurations, and they can vary substantially across the support.

More importantly, reach and curvature describe the geometry of a fixed set rather than the geometry of a probability distribution evolving under diffusion. A small region of extreme curvature carrying negligible probability mass can dominate a worst-case geometric bound despite having little relevance to the dynamics experienced by typical samples.

These observations suggest that the appropriate object should be neither purely topological nor purely worst-case geometric. We instead seek a quantity which is both geometric and probabilistic, and which changes continuously with the noise level.

1.4 Objects Investigated

We investigated several candidates before arriving at the posterior covariance field. Each captures a different aspect of the geometry of the diffusion process.

Pullback geometry of the flow. The derivative of the learned flow provides a natural metric describing how perturbations at time t are amplified or suppressed by the subsequent dynamics. This leads to the pullback metric

$$G_{t,T}(x) = D_x\hat{\Phi}_{t,T}(x)^\top D_x\hat{\Phi}_{t,T}(x).$$

This object turns out to be directly useful for understanding sampling stability. In section 3, we derive an anisotropic Wasserstein stability bound in which the velocity error is weighted according to the directions in which the learned dynamics subsequently amplify or contract it. However, $G_{t,T}$ describes the geometry of the learned dynamics, rather than the intrinsic geometry of the target distribution itself. Because we do not have access to the velocity error, this bound is not practical.

Level sets. Level sets of the smoothed density provide a classical geometric construction. Their tangent spaces and curvature are determined by the gradient and Hessian of p_t. Gaussian smoothing suppresses high-frequency components of these derivatives, giving a precise sense in which fine-scale geometric structure is progressively removed. Nevertheless, individual level sets can undergo topological transitions as the noise level changes, and their geometry depends on an arbitrary choice of level value. Therefore, we move on to the object around which the level sets concentrate.

Ridges. Density ridges provide a more direct attempt to identify the lower-dimensional structure around which probability mass concentrates. A ridge is defined by requiring the gradient to vanish in directions corresponding to sufficiently negative Hessian eigenvalues (normal directions). In the small-noise regime, ridges can serve as accurate surrogates for an underlying manifold; for example, Genovese et al. (2014) establish Hausdorff approximation and topological equivalence under suitable assumptions.

However, the ridge construction requires fixing an intrinsic dimension and depends discontinuously on spectral thresholding. When Hessian eigenvalues become degenerate, the associated normal eigenspaces are no longer uniquely defined. Similarly, changes in the sign of a Hessian eigenvalue can create or destroy components of the ridge. These issues become increasingly important away from the small-noise regime, precisely where we want a description of the geometry of diffusion.

Posterior covariance. The final object considered is the posterior covariance

$$A_t(y) := \frac{1}{t}\operatorname{Cov}(X \mid Y_t = y), \qquad Y_t = X + \sqrt{t}Z.$$

Unlike a manifold or ridge, $A_t(y)$ is defined for every $t > 0$ and every observation y. More importantly, it is directly connected to the local differential geometry of the score:

$$\nabla^2 \log p_t(y) = \frac{1}{t}\big(A_t(y) - I_D\big).$$

Thus the eigenvalues of A_t simultaneously describe posterior uncertainty and the local curvature of the log-density. Values close to 1 correspond to directions in which posterior uncertainty is comparable to the injected noise, while values close to 0 correspond to directions in which the noisy observation strongly identifies the underlying clean configuration.

This connection makes A_t a natural bridge between probability, geometry, and diffusion dynamics. Rather than forcing the distribution into a hard tangent/normal decomposition, it provides a soft spectral description of how many effective directions remain uncertain at each point and noise scale.

1.5 Main Contributions

The main contribution of this report is to propose the posterior covariance spectrum as a useful continuous description of geometric complexity throughout the diffusion process. The analysis has three principal components.

1. First, we derive an anisotropic stability bound for deterministic transport. The usual one-sided Lipschitz estimate controls all velocity errors using a single worst-case expansion rate. We show instead that the terminal effect of an error $\delta v_t(x)$ is determined by the pullback metric

$$G_{t,T}(x) = D_x\hat{\Phi}_{t,T}(x)^\top D_x\hat{\Phi}_{t,T}(x).$$

   This retains the directional structure of the learned dynamics and distinguishes errors which are subsequently amplified from those which are contracted.

2. Second, we investigate level sets and density ridges as candidate descriptions of the evolving geometry. These constructions clarify an important distinction between *geometric smoothing* and *continuous geometric evolution*. Gaussian convolution suppresses fine-scale spatial frequencies and decreases Sobolev energies monotonically, but it does not imply pointwise monotonicity of curvature or continuous evolution of ridge topology. This limits the usefulness of hard geometric objects at moderate and large noise scales.

3. Third, we introduce several scalar functionals of the posterior covariance spectrum. In particular, the effective dimension

$$d_{\text{eff}}(t) = \mathbb{E}\,\mathrm{tr}\,A_t(Y_t)$$

   provides a continuous analogue of intrinsic dimension. We supplement it with the distance to the set of orthogonal projectors,

$$\Delta_{\mathrm{proj}}(t) = \mathbb{E}\sum_i \min\{a_i^2, (a_i - 1)^2\},$$

which measures how closely the local posterior geometry resembles a genuine tangent-space projector, and the expansion functional

$$\mathcal{A}(t) = \mathbb{E}\sum_i (a_i - 1)_+^2,$$

which isolates directions in which posterior uncertainty exceeds the injected noise scale.

4. A further consequence is an adaptive discretisation principle. The evolution of the MMSE satisfies

$$\mathrm{MMSE}'(t) = \mathbb{E}\|A_t(Y_t)\|_F^2.$$

Combining this identity with the local discretisation error of the reverse process gives the step-size scaling

$$\Delta t(t) \propto \frac{t}{\sqrt{\mathbb{E}\|A_t(Y_t)\|_F^2}}.$$

Thus the same posterior covariance field which describes the evolving geometry also determines, through its spectral second moment, where a diffusion sampler should allocate computational effort.

Main Takeaway Our final perspective is that diffusion does not possess a single fixed intrinsic geometry. Instead, the geometry relevant to the dynamics is scale-dependent and probabilistic. At small noise, the posterior covariance approaches a tangent-space projector and recovers the usual manifold picture, under the manifold hypothesis. At larger noise, its spectrum becomes fractional and can contain expansive directions associated with posterior ambiguity. The posterior covariance field therefore provides a continuous interpolation between classical low-dimensional geometry and the full-dimensional probabilistic geometry of the noisy distribution.

2 Background

2.1 The Manifold Hypothesis

The manifold hypothesis offers an explanation for the effectiveness of machine learning algorithms on unstructured, high-dimensional data. It posits that a data distribution $X \sim p_{\text{data}}$ with $X \in \mathbb{R}^D$ lies on (or is concentrated near) a submanifold $\mathcal{M}$ of significantly lower dimension $d \ll D$. This restricts the effective degrees of freedom to d, enabling models to learn the underlying structure even in regimes where the sample size is smaller than the ambient dimension ($n < D$).

The primary motivation for the manifold hypothesis stems from the curse of dimensionality, wherein the volume of high-dimensional space grows exponentially, rendering data extremely sparse. High-dimensional spaces also exhibit counter-intuitive geometric behaviours; for instance, standard Gaussian data $\mathcal{N}(0, I_D)$ concentrates tightly within a thin spherical shell of radius $\sqrt{D}$.

The rationale behind the manifold hypothesis is that natural data distributions are typically locally continuous and governed by relatively few degrees of freedom. For instance, the space of realistic natural images constitutes a tiny fraction of the space of all possible pixel configurations.

Several variations of the manifold hypothesis relax its classical assumptions:

- **The Union of Manifolds Hypothesis** (Brown et al., 2022a,b): Argues that the classical single-manifold assumption is overly restrictive, modelling data instead as a union of multiple submanifolds of potentially varying dimensions.

- **The Stratified Space Hypothesis** (Aamari and Berenfeld, 2024): Extends this formulation further by permitting intersections between submanifolds of different dimensions.

For completeness, we review the formal definitions of a manifold and its reach.

Definition 2.1 (Manifold)**.** A d-dimensional topological manifold $\mathcal{M}$ is a topological space that is locally homeomorphic to an open subset of $\mathbb{R}^d$. Specifically, for each $y \in \mathcal{M}$, there exists an open neighbourhood $U_y \subseteq \mathcal{M}$ containing y and a continuous mapping $\Phi_y : U_y \to \mathbb{R}^d$ such that Φ_y is a homeomorphism onto its image. The smoothness of $\mathcal{M}$ is governed by the differentiability of the coordinate maps Φ_y.

When a manifold $\mathcal{M}$ is embedded in $\mathbb{R}^D$, its regularity can be quantified via the concept of *reach* (Federer, 1959):

Definition 2.2 (Reach)**.** The reach $\tau = \tau(\mathcal{M})$ of an embedded submanifold $\mathcal{M} \subset \mathbb{R}^D$ is defined as

$$\tau := \sup \{\varepsilon > 0 : \forall x \in \mathcal{M}^\varepsilon,\ \exists! y \in \mathcal{M} \text{ s.t. } \operatorname{dist}(x, \mathcal{M}) = \|x - y\|\},$$

where $\mathcal{M}^\varepsilon = \left\{x \in \mathbb{R}^D : \mathrm{dist}(x, \mathcal{M}) < \varepsilon\right\}$ denotes the open ε-neighbourhood of $\mathcal{M}$. Equivalently, τ is the supremum over all neighbourhood radii within which every point admits a unique orthogonal projection onto $\mathcal{M}$.

2.2 Flow Matching as a Score-Based Generative Model

Score-Based Generative Models (SGMs) (Ho et al., 2020; Song and Ermon, 2019; Song et al., 2020b) formulate generative modelling by estimating the score function, defined as the gradient of the log probability density $\nabla_x \log p(x)$. As noted by Song and Ermon (2019):

> "Because gradients can be ill-defined and hard to estimate when the data resides on **low-dimensional manifolds**, we perturb the data with different levels of Gaussian noise, and jointly estimate the corresponding scores, i.e., the vector fields of gradients of the perturbed data distribution for all noise levels."

Flow Matching (Lipman et al., 2022; Albergo et al., 2025) provides a continuous-time vector field framework that unifies and generalises SGMs. Below, we outline the connection between Gaussian continuous flows and score matching.

Gaussian Flows and Interpolation Schedule. In flow matching, the objective is to construct a continuous transport map pushing a base noise distribution $p_1 = \mathcal{N}(0, I_D)$ to a target data distribution p_0. Let $X_0 \sim p_0$ and $X_1 = \varepsilon \sim \mathcal{N}(0, I_D)$. The linear conditional path $X_t = \varphi_t(X_0, \varepsilon)$ interpolates between data and noise according to:

$$X_t = c_t X_0 + \sigma_t \varepsilon, \tag{1}$$

where $c_t, \sigma_t \in \mathbb{R}$ are continuously differentiable schedule functions satisfying boundary conditions $c_0 = 1, \sigma_0 = 0$ and $c_1 = 0, \sigma_1 = 1$. Differentiating X_t with respect to time t yields the conditional velocity field:

$$\dot{X}_t = \dot{c}_t X_0 + \dot{\sigma}_t \varepsilon. \tag{2}$$

Vector Field Formulation. The marginal probability density p_t of X_t evolves according to the continuity equation $\partial_t p_t = -\nabla \cdot (v_t p_t)$. The marginal vector field $v_t(x)$ generating this continuous density transformation is obtained via the conditional expectation:

$$v_t(x) = \mathbb{E}\left[\dot{X}_t \mid X_t = x\right] = \dot{c}_t \mathbb{E}\left[X_0 \mid X_t = x\right] + \dot{\sigma}_t \mathbb{E}\left[\varepsilon \mid X_t = x\right]. \tag{3}$$

Derivation via Tweedie's Formula. Since p_t corresponds to the distribution of $c_t X_0$ corrupted by additive Gaussian noise $\sigma_t \varepsilon \sim \mathcal{N}(0, \sigma_t^2 I_D)$, Tweedie's identity yields closed-form relationships between the conditional expectations and the score function $\nabla_x \log p_t(x)$:

$$\mathbb{E}\left[\varepsilon \mid X_t = x\right] = -\sigma_t \nabla_x \log p_t(x), \tag{4}$$

$$\mathbb{E}\left[X_0 \mid X_t = x\right] = \frac{x + \sigma_t^2 \nabla_x \log p_t(x)}{c_t}. \tag{5}$$

Substituting Equations (4) and (5) into the marginal velocity expression (3) gives:

$$\begin{aligned}
v_t(x) &= \dot{c}_t \left(\frac{x + \sigma_t^2 \nabla_x \log p_t(x)}{c_t} \right) + \dot{\sigma}_t \left(-\sigma_t \nabla_x \log p_t(x) \right) \\
&= \frac{\dot{c}_t}{c_t} x + \left(\frac{\dot{c}_t \sigma_t^2}{c_t} - \dot{\sigma}_t \sigma_t \right) \nabla_x \log p_t(x) \\
&= \frac{\dot{c}_t}{c_t} x + \sigma_t^2 \left(\frac{\dot{c}_t}{c_t} - \frac{\dot{\sigma}_t}{\sigma_t} \right) \nabla_x \log p_t(x).
\end{aligned} \tag{6}$$

Equivalence of Parameterisations. Equation (6) demonstrates that the velocity field $v_t(x)$ is uniquely determined by the noise schedules c_t, σ_t and the score function $\nabla_x \log p_t(x)$. Consequently, the following target parameterisations are equivalent up to deterministic pointwise linear transformations:

$$\nabla_x \log p_t(x) \quad \Longleftrightarrow \quad \mathbb{E}\left[\varepsilon \mid X_t = x\right] \quad \Longleftrightarrow \quad \mathbb{E}\left[X_0 \mid X_t = x\right] \quad \Longleftrightarrow \quad v_t(x). \tag{7}$$

Sampling via the deterministic flow ODE $\dot{x}_t = v_t(x_t)$ parameterised by a score model recovers the probability flow ODE framework (e.g., DDIM (Song et al., 2020a)).

2.3 Tweedie's 1st and 2nd Order Formulae

Consider the standard unscaled Gaussian perturbation model $X_t = X_0 + \sigma_t \varepsilon$, where $\varepsilon \sim \mathcal{N}(0, I_D)$ and $X_0 \sim p_0$. The conditional density of X_t given $X_0 = x_0$ is:

$$p_t(x \mid x_0) = (2\pi\sigma_t^2)^{-D/2} \exp\left(-\frac{\|x - x_0\|^2}{2\sigma_t^2} \right). \tag{8}$$

The marginal density $p_t(x)$ is given by:

$$p_t(x) = \int_{\mathbb{R}^D} p_t(x \mid x_0) p_0(x_0)\, dx_0. \tag{9}$$

2.3.1 First Derivative (Score Function)

Differentiating $p_t(x)$ under the integral sign with respect to x:

$$\nabla_x p_t(x) = -\frac{1}{\sigma_t^2} \int_{\mathbb{R}^D} (x - x_0) p_t(x \mid x_0) p_0(x_0)\, dx_0. \tag{10}$$

Dividing by $p_t(x)$ and applying Bayes' rule $p(x_0 \mid x) = \frac{p_t(x|x_0)p_0(x_0)}{p_t(x)}$ yields Tweedie's first-order identity:

$$\nabla_x \log p_t(x) = \frac{\nabla_x p_t(x)}{p_t(x)} = \frac{\mathbb{E}[X_0 \mid X_t = x] - x}{\sigma_t^2}. \tag{11}$$

2.3.2 Second Derivative (Hessian Matrix / Score Jacobian)

Expanding the Hessian matrix of the log-marginal density:

$$\nabla_x^2 \log p_t(x) = \frac{\nabla_x^2 p_t(x)}{p_t(x)} - \left(\nabla_x \log p_t(x)\right)\left(\nabla_x \log p_t(x)\right)^T . \tag{12}$$

Differentiating $\nabla_x p_t(x)$ a second time yields:

$$\nabla_x^2 p_t(x) = \int_{\mathbb{R}^D} \left(-\frac{1}{\sigma_t^2} I_D + \frac{(x-x_0)(x-x_0)^T}{\sigma_t^4} \right) p_t(x \mid x_0) p_0(x_0)\, dx_0. \tag{13}$$

Dividing by $p_t(x)$:

$$\frac{\nabla_x^2 p_t(x)}{p_t(x)} = -\frac{1}{\sigma_t^2} I_D + \frac{1}{\sigma_t^4} \mathbb{E}\left[(x-X_0)(x-X_0)^T \mid X_t = x\right]. \tag{14}$$

Substituting Equations (11) and (14) into Equation (12):

$$\begin{aligned}\nabla_x^2 \log p_t(x) = -\frac{1}{\sigma_t^2} I_D + \frac{1}{\sigma_t^4} \Big(&\mathbb{E}\left[(x-X_0)(x-X_0)^T \mid X_t = x\right] \\ &- \mathbb{E}\left[x - X_0 \mid X_t = x\right] \mathbb{E}\left[x - X_0 \mid X_t = x\right]^T \Big).\end{aligned} \tag{15}$$

Because x is deterministic given $X_t = x$, the bracketed term reduces to the conditional posterior covariance $\mathrm{Cov}(X_0 \mid X_t = x)$. Thus, Tweedie's second-order formula simplifies to:

$$\nabla_x^2 \log p_t(x) = \frac{1}{\sigma_t^4} \mathrm{Cov}(X_0 \mid X_t = x) - \frac{1}{\sigma_t^2} I_D. \tag{16}$$

This result is extremely helpful, and not at all obvious. It allows us to compute posterior covariances by autodifferentiation of a score network.

Takeaway.

- The manifold hypothesis broadly states that data lie on lower-dimensional submanifold(s) of a higher-dimensional Euclidean space,
- From the gradient of the log likelihood of the noised density we can sample from a target density by (numerically) solving a vector field ODE to transport a known density to a target density,
- Tweedie's 1^{st} and 2^{nd} order formulae relate the posterior mean and posterior covariance to the score and the score Jacobian, respectively.

With this background we can now explore the geometry of diffusion.

3 Anisotropic Stability Bound

We record a stability estimate for generator matching that tracks *which* directions of a drift error are amplified and which are dampened. The scalar one-sided Lipschitz estimate propagates every error with a single worst-case rate, making it sharp only in an isotropic sense. The estimate below replaces that scalar rate with the derivative flow of the learned dynamics.

3.1 The Deterministic Transport Mechanism

Consider two time-dependent velocity fields $v_t, \hat{v}_t : \mathbb{R}^D \to \mathbb{R}^D$ for $0 \le t \le T$, smooth enough in space and integrable in time such that their characteristic flows are globally defined and differentiable with respect to the initial condition:

$$\tfrac{\mathrm{d}}{\mathrm{d}s}\Phi_{t,s}(x) = v_s\big(\Phi_{t,s}(x)\big), \qquad \tfrac{\mathrm{d}}{\mathrm{d}s}\hat{\Phi}_{t,s}(x) = \hat{v}_s\big(\hat{\Phi}_{t,s}(x)\big), \qquad \Phi_{t,t} = \hat{\Phi}_{t,t} = \mathrm{Id}. \tag{17}$$

We write $\delta v_t := v_t - \hat{v}_t$ for the velocity error, where v_t represents the exact field and $\hat{v}_t$ the learned field.

The estimate is constructed using the derivative of the *learned* flow, which is accessible directly from its neural network architecture:

$$\hat{J}_{t,T}(x) := D_x\hat{\Phi}_{t,T}(x), \qquad \tfrac{\mathrm{d}}{\mathrm{d}s}\hat{J}_{t,s}(x) = \nabla\hat{v}_s\big(\hat{\Phi}_{t,s}(x)\big)\,\hat{J}_{t,s}(x), \qquad \hat{J}_{t,t}(x) = \mathrm{Id}. \tag{18}$$

Thus, $\hat{J}_{t,T}(x)h$ denotes the terminal displacement at time T produced by an infinitesimal perturbation h applied at (t,x) and transported by the learned dynamics. An error at time t is not rescaled by a uniform worst-case constant; rather, it is carried along its specific direction by $\hat{J}_{t,T}$. The following identity formalises this mechanism.

Lemma 3.1 (Telescoping identity along characteristics)**.** *For every initial point $x_0 \in \mathbb{R}^D$, let $X_t := \Phi_{0,t}(x_0)$ denote the exact characteristic trajectory. Then,*

$$\Phi_{0,T}(x_0) - \hat{\Phi}_{0,T}(x_0) = \int_0^T \hat{J}_{t,T}(X_t)\,\delta v_t(X_t)\,\mathrm{d}t. \tag{19}$$

Proof. Define $F(t) := \hat{\Phi}_{t,T}(X_t)$, which tracks the exact dynamics up to time t and transitions to the learned dynamics from t to T. Its boundary values represent the fully learned and fully exact terminal points: $F(0) = \hat{\Phi}_{0,T}(x_0)$ and $F(T) = \Phi_{0,T}(x_0)$. Differentiating $F(t)$ with respect to t yields:

$$\tfrac{\mathrm{d}}{\mathrm{d}t}F(t) = \partial_t\hat{\Phi}_{t,T}(X_t) + D\hat{\Phi}_{t,T}(X_t)\,\dot{X}_t.$$

By initiating the learned flow an instant later, we eliminate an infinitesimal learned step, giving $\partial_t\hat{\Phi}_{t,T}(x) = -D\hat{\Phi}_{t,T}(x)\,\hat{v}_t(x)$. Combining this with $\dot{X}_t = v_t(X_t)$ and $D\hat{\Phi}_{t,T}(X_t) = \hat{J}_{t,T}(X_t)$, we obtain:

$$\tfrac{\mathrm{d}}{\mathrm{d}t}F(t) = \hat{J}_{t,T}(X_t)\big(v_t(X_t) - \hat{v}_t(X_t)\big) = \hat{J}_{t,T}(X_t)\,\delta v_t(X_t).$$

Integrating $\tfrac{\mathrm{d}}{\mathrm{d}t}F(t)$ from $t=0$ to $t=T$ proves (19). ∎

Identity (19) shows that the local velocity error at each instant enters the terminal gap through the derivative flow evaluated precisely along the actual error direction, avoiding worst-case scalar overestimation.

3.2 From the Telescoping Identity to the Matrix Bound

We now convert this pathwise identity into a bound on the target probability measures via optimal transport coupling. Let $X_0 \sim \rho_0$ and transport it using both velocity fields: $X_t := \Phi_{0,t}(X_0)$ and $\hat{X}_t := \hat{\Phi}_{0,t}(X_0)$, yielding terminal laws $\rho_T := \mathrm{Law}(X_T)$ and $\hat{\rho}_T := \mathrm{Law}(\hat{X}_T)$. Driving both flows from the same random seed X_0 forms a valid coupling between ρ_T and $\hat{\rho}_T$, giving $W_1(\rho_T, \hat{\rho}_T) \leq \mathbb{E}\|X_T - \hat{X}_T\|$. Applying Lemma 3.1 pathwise at $x_0 = X_0$ alongside the integral triangle inequality yields:

$$W_1(\rho_T, \hat{\rho}_T) \leq \int_0^T \mathbb{E}\big\|\hat{J}_{t,T}(X_t)\,\delta v_t(X_t)\big\|\,\mathrm{d}t = \int_0^T \int_{\mathbb{R}^D} \big\|\hat{J}_{t,T}(x)\,\delta v_t(x)\big\|\,\rho_t(\,\mathrm{d}x)\,\mathrm{d}t, \tag{20}$$

where the final equality follows from $X_t \sim \rho_t$. It remains to express the matrix norm via its underlying quadratic form.

Theorem 3.2 (Anisotropic W_1 stability)**.** *Define the pullback amplification metric tensor:*

$$G_{t,T}(x) := \hat{J}_{t,T}(x)^\top \hat{J}_{t,T}(x), \qquad \text{such that} \qquad \|\hat{J}_{t,T}(x)h\|^2 = \big\langle h,\, G_{t,T}(x)\,h\big\rangle. \tag{21}$$

Then the Wasserstein distance satisfies:

$$W_1(\rho_T, \hat{\rho}_T) \leq \int_0^T \int_{\mathbb{R}^D} \big\langle \delta v_t(x),\, G_{t,T}(x)\,\delta v_t(x)\big\rangle^{1/2}\,\rho_t(\,\mathrm{d}x)\,\mathrm{d}t. \tag{22}$$

Proof. Substituting the norm identity (21) directly into (20) completes the proof. $\square$

The matrix $G_{t,T}(x)$ represents the standard Euclidean metric at terminal time T pulled back to the tangent space at (t,x) via the learned flow. It operates at time t while measuring vector magnitudes following transport to time T. Its eigenvectors identify principal directionalities at (t,x), and its eigenvalues correspond to squared directional stretching factors. Bound (22) is anisotropic because it explicitly evaluates $G_{t,T}$ along the error vector $\delta v_t(x)$ rather than maximising over all directions.

3.3 Recovering the One-Sided Lipschitz Estimate

The classical scalar stability estimate represents the isotropic limit of Theorem 3.2, obtained by discarding directional information. Let $S_t(x) := \frac{1}{2}\big(\nabla \hat{v}_t(x) + \nabla \hat{v}_t(x)^\top\big)$ be the symmetric Jacobian component. Assume the learned field is one-sided Lipschitz with rate ℓ_t, namely:

$$\lambda_{\max}\big(S_t(x)\big) \leq \ell_t \qquad \text{for all } x \in \mathbb{R}^D. \tag{23}$$

This condition restricts directional expansion by capping the maximum separation rate between trajectories, without constraining local contractions or rotational shearing in $\nabla\hat{v}_t$.

Corollary 3.3 (Scalar one-sided Lipschitz bound)**.** *Under assumption* (23),

$$G_{t,T}(x) \preceq \exp\left(2\int_t^T \ell_s\,\mathrm{d}s\right)\mathrm{Id}, \tag{24}$$

which implies the scalar stability bound:

$$W_1(\rho_T, \hat{\rho}_T) \leq \int_0^T \exp\left(\int_t^T \ell_s\,\mathrm{d}s\right)\|\delta v_t\|_{L^1(\rho_t)}\,\mathrm{d}t. \tag{25}$$

Proof. Fix a direction vector h and set $z_s := \hat{J}_{t,s}(x)h$, which satisfies $\dot{z}_s = \nabla\hat{v}_s(\hat{\Phi}_{t,s}(x))z_s$ according to (18). Only the symmetric Jacobian component drives norm growth:

$$\tfrac{\mathrm{d}}{\mathrm{d}s}\|z_s\|^2 = 2\big\langle z_s,\, S_s(\hat{\Phi}_{t,s}(x))z_s\big\rangle \leq 2\ell_s\|z_s\|^2.$$

Applying Grönwall's inequality yields $\|\hat{J}_{t,T}(x)h\| \leq \exp\Big(\int_t^T \ell_s\,\mathrm{d}s\Big)\|h\|$, establishing matrix inequality (24). Substituting (24) into (22) and using $\langle\delta v_t, G_{t,T}\delta v_t\rangle^{1/2} \leq \exp\Big(\int_t^T \ell_s\,\mathrm{d}s\Big)\|\delta v_t\|$ concludes the proof. □

Corollary 3.3 illustrates that the scalar estimate relies on the Loewner order bound (24): $G_{t,T}$ is uniformly dominated by its largest eigenvalue across all directions. While this bound holds generally, it ignores the eigenstructure of $G_{t,T}$, treating strongly contracting error directions identically to maximal expansion directions. This overestimation is especially pronounced when velocity errors lie normal to a low-dimensional target manifold.

3.4 A Two-Direction Model

The gap between matrix and scalar estimates can be observed in a simple two-direction model. Consider an orthogonal decomposition $\mathbb{R}^D = E_\parallel \oplus E_\perp$ into tangent-like and normal-like subspaces. Writing $x = (x_\parallel, x_\perp)$, assume a linear, block-diagonal learned velocity field that expands along $E_\parallel$ and contracts along $E_\perp$:

$$\hat{v}_t(x_\parallel, x_\perp) = \big(\ell_t x_\parallel,\, -\gamma_t x_\perp\big), \qquad \gamma_t \geq 0. \tag{26}$$

Its spatial Jacobian $\nabla\hat{v}_t = \mathrm{diag}(\ell_t I_\parallel, -\gamma_t I_\perp)$ is constant in space. Consequently, the derivative ODE (18) integrates blockwise, yielding diagonal expressions for $\hat{J}_{t,T}$ and $G_{t,T}$:

$$\hat{J}_{t,T} = \begin{pmatrix} e^{\int_t^T \ell_s\,\mathrm{d}s} I_\parallel & 0 \\ 0 & e^{-\int_t^T \gamma_s\,\mathrm{d}s} I_\perp \end{pmatrix}, \qquad G_{t,T} = \begin{pmatrix} e^{2\int_t^T \ell_s\,\mathrm{d}s} I_\parallel & 0 \\ 0 & e^{-2\int_t^T \gamma_s\,\mathrm{d}s} I_\perp \end{pmatrix}. \tag{27}$$

Decomposing the drift error as $\delta v_t = \delta v_t^{\parallel} + \delta v_t^{\perp}$ with $\delta v_t^{\parallel} \in E_{\parallel}$ and $\delta v_t^{\perp} \in E_{\perp}$, the matrix bound integrand in (22) separates into:

$$\left\langle \delta v_t,\, G_{t,T}\, \delta v_t \right\rangle^{1/2} = \left(e^{2\int_t^T \ell_s\, \mathrm{d}s} \|\delta v_t^{\parallel}\|^2 + e^{-2\int_t^T \gamma_s\, \mathrm{d}s} \|\delta v_t^{\perp}\|^2 \right)^{1/2}. \tag{28}$$

Tangential errors are propagated to terminal time via expansion factor $e^{\int_t^T \ell}$, whereas normal errors are suppressed by decay factor $e^{-\int_t^T \gamma}$.

The scalar estimate cannot separate these components. Its expansion rate corresponds to the maximum eigenvalue of the symmetric Jacobian:

$$\lambda_{\max}\left(S_t\right) = \max\{\ell_t,\, -\gamma_t\} = \ell_t \qquad (\ell_t \geq 0), \tag{29}$$

forcing Corollary 3.3 to propagate *both* error components using the tangential expansion factor $e^{\int_t^T \ell}$. By replacing normal contraction $e^{-\int_t^T \gamma}$ with expansion, it overestimates normal error contributions by a factor of $e^{\int_t^T (\ell_s + \gamma_s)\, \mathrm{d}s}$. In regimes near target manifolds where contraction is strong ($\gamma_t \sim (T-t)^{-1}$, making $e^{-\int_t^T \gamma}$ decay polynomially in $T-t$), this overestimate creates the illusion of persistent error accumulation where damping actually occurs.

3.5 Structural Requirements on the Drift Error

The matrix and scalar bounds require different levels of information about the velocity error. The scalar estimate (25) depends only on the spatial norm $\|\delta v_t\|_{L^1(\rho_t)}$. Conversely, the matrix bound (22) depends on the directional quadratic form $\langle \delta v_t, G_{t,T} \delta v_t \rangle$, requiring knowledge of the orientation of δv_t relative to the eigenframe of $G_{t,T}$. Because the ground-truth velocity v_t is unknown in practice, $\delta v_t = v_t - \hat{v}_t$ cannot be directly evaluated.

Although empirical observations suggest that δv_t often aligns with v_t or exhibits Gaussian fluctuations across random training runs, the statistical behaviour of δv_t is generally unknown prior to training.

To construct a practical coordinate system, we define fixed projection operators (P_t onto the tangential subspace and $Q_t = I - P_t$ onto the normal subspace) generated exclusively by the geometry of the target distribution. Re-estimating the eigenframe for each learned model individually would introduce model-dependent weights and complicate comparative evaluations. Therefore, we fix P_t once using the target density geometry, measuring all model errors within this single benchmark frame. The projection P_t is derived directly from the spectral decomposition of the posterior-covariance operator A_t detailed in section 5.

Conclusion on Anisotropic Stability. Drift errors δv_t propagate through the learned flow's derivative $\hat{J}_{t,T}$; error amplification or dampening depends on the specific spatial direction of the error rather than a single global scale. Classical bounds over-estimate this error by treating strongly contracting directions the same as expanding ones. This is important in the manifold picture where contracting dynamics along normal directions to a target manifold actively suppress normal drift errors.

4 Level Sets and Ridges

4.1 Level Sets

As we smooth data lying on (or concentrated around) a manifold, two main effects occur: smoothing in the normal direction and smoothing in the tangent direction. Smoothing in the tangent direction makes a density more uniform along the manifold, leading to better-behaved level sets. Here, we investigate how these level sets evolve under smoothing.

Let $p^\star : \mathbb{R}^D \to \mathbb{R}_+$ denote the initial density, and let p_σ denote its Gaussian-smoothed counterpart:

$$p_\sigma(x) = (G_\sigma * p^\star)(x), \qquad G_\sigma(x) = \frac{1}{(2\pi\sigma^2)^{D/2}} \exp\left(-\frac{\|x\|^2}{2\sigma^2}\right). \tag{30}$$

For a regular value c, the corresponding level set is

$$\mathcal{M}_c^\sigma := \left\{x \in \mathbb{R}^D : p_\sigma(x) = c\right\}, \qquad \nabla p_\sigma(x) \neq 0 \quad \text{for } x \in \mathcal{M}_c^\sigma. \tag{31}$$

The unit normal vector field to $\mathcal{M}_c^\sigma$ is

$$n(x) = \frac{\nabla p_\sigma(x)}{\|\nabla p_\sigma(x)\|}. \tag{32}$$

For a tangent vector $v \in T_x\mathcal{M}_c^\sigma$, the directional derivative of the unit normal is

$$D_v n = \frac{1}{\|\nabla p_\sigma\|} \left(\mathrm{Id} - nn^\top\right) D^2 p_\sigma\, v = \frac{1}{\|\nabla p_\sigma\|} P_T D^2 p_\sigma\, v, \tag{33}$$

where

$$P_T := \mathrm{Id} - nn^\top \tag{34}$$

denotes the orthogonal projection onto the tangent space $T_x\mathcal{M}_c^\sigma$. Thus, the variation of the normal vector field—and consequently the local curvature of the level set—is determined by the Hessian of the smoothed density.

Gaussian smoothing regularises not only the density itself, but all of its spatial derivatives. In particular, for any $k \geq 0$,

$$D^k p_\sigma = D^k(G_\sigma * p^\star) = (D^k G_\sigma) * p^\star. \tag{35}$$

Hence, increasing σ smooths the density while simultaneously suppressing fine-scale variations in its gradient, Hessian, and higher-order derivatives.

This smoothing has a straightforward interpretation in the Fourier domain. Let $\widehat{p}_\sigma$ denote the Fourier transform:

$$\widehat{p}_\sigma(\xi) = \widehat{G}_\sigma(\xi)\, \widehat{p^\star}(\xi) = \exp\left(-\frac{\sigma^2\|\xi\|^2}{2}\right) \widehat{p^\star}(\xi). \tag{36}$$

A Fourier mode of spatial frequency $\|\xi\|$ is attenuated by $\exp(-\sigma^2\|\xi\|^2/2)$. Because high-frequency components decay exponentially with $\|\xi\|^2$, Gaussian smoothing preferentially eliminates the fine-scale spatial variations of $p^\star$.

The effect on the geometry of a level set follows directly from (33). Up to a choice of orientation for the normal vector, its shape operator is

$$S := -P_T D n P_T = -\frac{1}{\|\nabla p_\sigma\|} P_T D^2 p_\sigma P_T. \tag{37}$$

The eigenvalues of S correspond to the principal curvatures of $\mathcal{M}_c^\sigma$. Therefore, Gaussian smoothing influences level-set geometry through two main factors: the Hessian $D^2 p_\sigma$, which controls normal variation, and $\|\nabla p_\sigma\|$, which scales density variation relative to geometric variation. Because Gaussian smoothing suppresses high-frequency components of $D^2 p_\sigma$, it dampens fine-scale variations of the shape operator and removes small-scale geometric structures from the level set.

It is important to distinguish this suppression of fine-scale variation from a uniform, pointwise reduction in curvature. In general, Gaussian smoothing does not guarantee that the principal curvatures of a fixed level set decrease monotonically with σ. Indeed, the shape operator

$$S_\sigma(x) = -\frac{1}{\|\nabla p_\sigma(x)\|} P_T D^2 p_\sigma(x) P_T, \qquad x \in \mathcal{M}_c^\sigma, \tag{38}$$

depends simultaneously on $D^2 p_\sigma$ and ∇p_σ, and is evaluated along the evolving level set $\mathcal{M}_c^\sigma$. Although Gaussian smoothing attenuates high-frequency modes of $D^2 p_\sigma$, it also alters ∇p_σ and the spatial location of the level set itself. Consequently, S_σ and its eigenvalues do not satisfy a general pointwise monotonicity property.

Furthermore, the topology of $\mathcal{M}_c^\sigma$ can change as σ varies. Such topological transitions occur when the level value c passes through a critical point where

$$p_\sigma(x) = c, \qquad \nabla p_\sigma(x) = 0. \tag{39}$$

For instance, two distinct connected components of a super-level set may merge through a neck as σ increases. Thus, while smoothing removes high-frequency spatial variations from the density and its derivatives, it should not be viewed as causing a monotone decrease in curvature for every individual level set. Rather, Gaussian smoothing preferentially suppresses spatial frequencies at scales smaller than σ, thereby filtering out fine-scale geometric structures encoded by the higher-order derivatives of the density.

More generally, adding Gaussian noise monotonically suppresses all L^2 Sobolev energies of the smoothed density. Let

$$p_\sigma = G_\sigma * p^\star, \qquad G_\sigma(x) = \frac{1}{(2\pi\sigma^2)^{D/2}} \exp\left(-\frac{\|x\|^2}{2\sigma^2}\right), \tag{40}$$

where D denotes the ambient dimension. Since p_σ satisfies the heat equation $\partial_{\sigma^2} p_\sigma = \frac{1}{2}\Delta p_\sigma$, we have, for every integer $k \geq 0$,

$$\frac{\partial}{\partial \sigma^2} \frac{1}{2} \int_{\mathbb{R}^D} \left| D^k p_\sigma(x) \right|^2 \mathrm{d}x = -\frac{1}{2} \int_{\mathbb{R}^D} \left| \nabla D^k p_\sigma(x) \right|^2 \mathrm{d}x \leq 0. \tag{41}$$

Increasing the noise level σ monotonically decreases the L^2 norms of all spatial derivatives of p_σ. In particular,

$$\frac{\partial}{\partial\sigma^2}\frac{1}{2}\int_{\mathbb{R}^D}|\nabla p_\sigma|^2\,\mathrm{d}x = -\frac{1}{2}\int_{\mathbb{R}^D}|\Delta p_\sigma|^2\,\mathrm{d}x \le 0, \tag{42}$$

$$\frac{\partial}{\partial\sigma^2}\frac{1}{2}\int_{\mathbb{R}^D}|D^2 p_\sigma|^2\,\mathrm{d}x = -\frac{1}{2}\int_{\mathbb{R}^D}|\nabla\Delta p_\sigma|^2\,\mathrm{d}x \le 0. \tag{43}$$

Equivalently, in the Fourier domain:

$$\frac{1}{2}\int_{\mathbb{R}^D}|D^k p_\sigma(x)|^2\,\mathrm{d}x \;\propto\; \frac{1}{2}\int_{\mathbb{R}^D}\|\xi\|^{2k}e^{-\sigma^2\|\xi\|^2}\left|\widehat{p^\star}(\xi)\right|^2\,\mathrm{d}\xi. \tag{44}$$

Increasing σ progressively filters out fine-scale features, suppressing spatial frequencies $\|\xi\|$ according to $e^{-\sigma^2\|\xi\|^2}$.

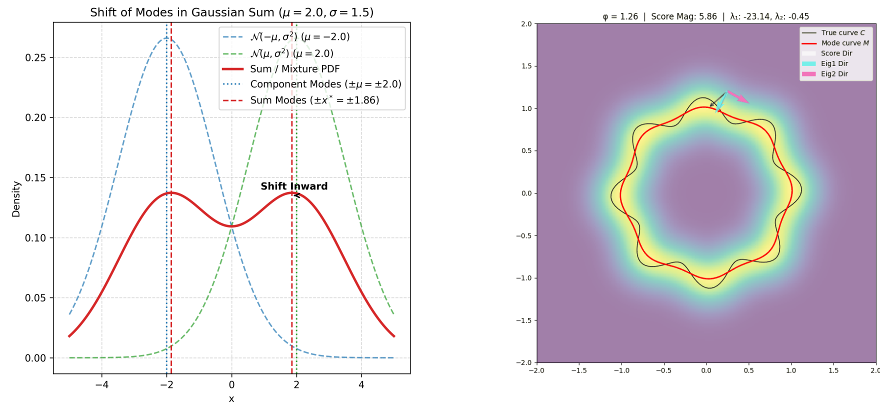
This behaviour aligns with numerical intuition in generative modelling and sampling, where finer time steps are required near the final data distribution ($t \to 0$ or $\sigma \to 0$) to resolve higher-frequency density details. This phenomenon is especially prominent in high-dimensional image and audio signals. Analogously, JPEG compression exploits Fourier-domain (or discrete cosine) representations by discarding high-frequency components that contribute negligibly to human visual perception.

Conclusion on Level Sets. Analysing the level sets of a density helps to explain how the higher frequency components decay quadratically faster than the lower frequency components. They are, however, not a tractable or indeed truly helpful tool for analysing the geometrical properties of a density, since they are discontinuous. Note that one would need to make a choice of level set and there is no canonical correct choice; picking a particular quantile is arbitrary. Since the level sets seem to contract around the initial support $\mathcal{M}$ for small amounts of added noise, we can instead analyse the object the level sets seem to be centred around. This motivates defining and looking at ridges of a density.

4.2 Motivation for the Ridge

An initially counterintuitive aspect of the manifold hypothesis is that the lower-dimensional space around which perturbed data concentrates is typically not the original manifold $\mathcal{M}$. As the noise level increases, the effective kernel bandwidth grows, encompassing a broader region of the data space. Consequently, the noise-corrupted density concentrates around an evolving low-dimensional structure. This raises the question: *What is this evolving space, and how does its geometry behave over time?* This key question in trying to understand the geometry of diffusion is what we try to answer here.

The phenomenon of concentrating around a new space is evident even when considering the convolution of simple discrete distributions, such as a mixture of Gaussian-smoothed point masses: the modes of the smoothed distribution do not coincide with the original support.



(a) Gaussian smoothing shifts modes in 1D

(b) Gaussian smoothing shifts the ridge in 2D

Figure 1: Comparison of ridge structures under varying smoothing scales.

With this visual intuition, we can formalise the definition of a ridge: it is the locus of points where the gradient of the density restricted to the normal directions vanishes. However, a priori, the normal directions are not fixed externally.

The key observation is to let the density determine its own normal frame. At a point on a d-dimensional ridge, the density exhibits ridge-like behaviour: it achieves a local maximum along the $D - d$ transverse directions, while varying comparatively smoothly along the remaining d intrinsic directions. Transverse to the ridge, the density curves sharply downward; along the ridge, its curvature is milder.

Mathematically, this condition is governed by the Hessian of the density. Let $g(x) = \nabla p(x)$ and $H(x) = \nabla^2 p(x)$, and denote the eigenvalues of $H(x)$ in descending order:

$$\lambda_1(x) \geq \lambda_2(x) \geq \cdots \geq \lambda_d(x) \geq \lambda_{d+1}(x) \geq \cdots \geq \lambda_D(x), \tag{45}$$

with corresponding orthonormal eigenvectors $u_1(x), \ldots, u_D(x)$. The directions of sharpest downward concavity correspond to the most negative eigenvalues. Thus, we define the normal space at x as the span of the bottom $D - d$ eigenvectors, and the tangent space as the span of the top d eigenvectors. Writing $V(x) = [u_{d+1}(x) \mid \cdots \mid u_D(x)]$ for the normal frame, the orthogonal projector onto the normal subspace is given by:

$$L(x) = V(x)V(x)^\top. \tag{46}$$

Importantly, the normal space is not specified exogenously; it is determined intrinsically by the second-order geometric structure of the density field itself.

A point x belongs to the d-dimensional ridge if the density cannot be increased by moving along any normal direction. Equivalently, the projection of the gradient onto the normal space vanishes. Defining the projected gradient:

$$G(x) := L(x)g(x), \tag{47}$$

the d-dimensional ridge set R is defined as:

$$R := \{x \in \mathbb{R}^D : \|G(x)\| = 0,\ \lambda_{d+1}(x) < 0\}. \tag{48}$$

The strict inequality $\lambda_{d+1}(x) < 0$ guarantees that even the largest eigenvalue in the normal direction is strictly negative, ensuring local maximality transverse to the ridge rather than a saddle point.

When $d = 0$, this definition recovers classical local modes: $L(x) = \mathrm{Id}$, the condition $\|G(x)\| = 0$ simplifies to $\nabla p(x) = 0$, and $\lambda_1(x) < 0$ isolates local maxima. A general d-ridge relaxes this requirement by demanding critical point behaviour solely within the $(D - d)$-dimensional normal subspace, coupled with negative concavity across the normal block.

4.3 Concentration of the Score along the Ridge

We first highlight a useful structural property: the ridge of $\log p_\sigma$ and the ridge of p_σ define the identical spatial set. Because the logarithmic function is strictly monotonic, $\nabla \log p_\sigma = p_\sigma^{-1} \nabla p_\sigma$ vanishes if and only if $\nabla p_\sigma = 0$. Furthermore, their Hessians satisfy:

$$\nabla^2 \log p_\sigma = \frac{1}{p_\sigma} \nabla^2 p_\sigma - \frac{1}{p_\sigma^2} \nabla p_\sigma \nabla p_\sigma^\top, \tag{49}$$

which coincides with $p_\sigma^{-1} \nabla^2 p_\sigma$ whenever the projected gradient vanishes on the ridge.

We work primarily with the log-density $\log p_\sigma$ for two key reasons:

1. Its gradient $\nabla \log p_\sigma$ constitutes the score function, which score-based generative models directly learn.

2. As established by Genovese et al. (2014), the surrogate approximation property (Theorem 4.1) holds within an $O(1)$ neighbourhood of $\mathcal{M}$ for the log-density, whereas it holds only within a restricted $O(\sigma)$ neighbourhood for the raw density p_σ.

Theorem 4.1 (Surrogate Ridge Approximation, Theorem 7 of (Genovese et al., 2014))**.** *Consider the Hidden Manifold Model with noise variance σ^2, and assume that the underlying manifold $\mathcal{M}$ is compact and boundaryless. Then the hyper-ridge set R_σ of the density p_σ satisfies:*

$$\mathrm{Haus}(\mathcal{M}, R_\sigma) = O\left(\sigma^2 \log \frac{1}{\sigma}\right), \tag{50}$$

and R_σ is topologically homeomorphic to $\mathcal{M}$. Consequently, in the small-noise limit $\sigma \to 0$ with $\mathcal{M}$ fixed, the ridge R_σ converges to $\mathcal{M}$ in Hausdorff distance.

Within an $O(\sigma)$ neighbourhood of the manifold, the score operates predominantly as a normal restoring force field. Genovese et al. (2014) showed that:

$$\nabla \log p_\sigma(x) = -\frac{1}{\sigma^2}\left((x-\hat{x}) + O(K_\sigma^2)\right), \tag{51}$$

where $\hat{x} = \pi_{\mathcal{M}}(x)$ denotes the orthogonal projection onto $\mathcal{M}$, and

$$K_\sigma = \sqrt{2\sigma^2 \log\left(\sigma^{-(A+D)}\right)} = O\left(\sigma\sqrt{\log \tfrac{1}{\sigma}}\right). \tag{52}$$

The normal component of the score has magnitude $O(\sigma^{-1})$, whereas the tangential component remains $O(1)$. Thus, as $\sigma \to 0$, the score becomes a strong linear restoring force toward the ridge rather than toward $\mathcal{M}$ directly.

This distinction becomes clear by comparing first-order Taylor expansions of the score around the manifold projection $m = \pi_{\mathcal{M}}(x)$ and around the ridge projection $r = \pi_{R_\sigma}(x)$:

$$\nabla \log p_\sigma(x) = \nabla_{\mathcal{M}} \log p_\sigma(m) - \frac{1}{\sigma^2}(x-m) + O(K_\sigma^2) + o\left(\frac{\|x-m\|}{\sigma^2}\right), \tag{53}$$

$$\nabla \log p_\sigma(x) = \nabla_{R_\sigma} \log p_\sigma(r) - \frac{1}{\sigma^2}(x-r) + o\left(\frac{\|x-r\|}{\sigma^2}\right). \tag{54}$$

Expanding around the manifold leaves a residual offset of order $O(K_\sigma^2)$, whereas expanding around the ridge eliminates this term. This residual acts as a constant normal bias of magnitude $K_\sigma^2/\sigma^2 = O(\log \frac{1}{\sigma})$, which persists on the manifold $\mathcal{M}$ itself. The zero-crossing of the normal restoring field is displaced from $\mathcal{M}$ by $O(K_\sigma^2)$ along the normal direction—a displacement matching the Hausdorff distance $\mathrm{Haus}(\mathcal{M}, R_\sigma) = O(\sigma^2 \log \frac{1}{\sigma})$ in Theorem 4.1.

Projecting (54) onto the normal subspace N_r at the ridge point r formalises this behaviour. Because the intrinsic gradient $\nabla_{R_\sigma} \log p_\sigma(r)$ lies entirely within the tangent space $T_r R_\sigma$, it is annihilated by the normal projector $P_N(r)$, yielding:

$$P_N(r)\nabla \log p_\sigma(x) = -\frac{1}{\sigma^2}(x-r) + o\left(\frac{\|x-r\|}{\sigma^2}\right). \tag{55}$$

Thus, the normal score component acts as a pure linear restoring force directed toward the ridge, with universal spring constant $-\sigma^{-2}$ and no curvature-induced residual. This confirms that the ridge set constitutes the precise geometric attractor of the score function's dominant restoring field.

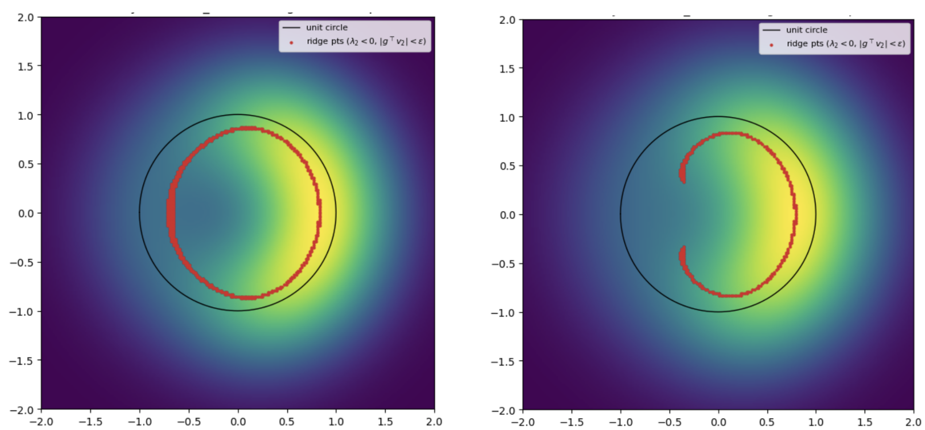
4.4 Limitations of the Ridge Formulation

The underlying density fields used to define the ridge are mathematically smooth. For any $t > 0$, the noise-smoothed density p_t is a Gaussian convolution and is therefore real-analytic. Consequently, the posterior mean $m_t(y)$, the posterior covariance field $A_t(y)$, and the score Hessian $\nabla^2 \log p_t(y) = \frac{1}{t}(A_t(y) - \mathrm{Id})$ are all analytic in y. Their eigenvalues $\lambda_1(y) \geq \cdots \geq \lambda_D(y)$ vary continuously throughout $\mathbb{R}^D$ (section 5).

However, despite the smoothness of p_t, the ridge set R_t itself does not depend continuously on the data distribution. Two distinct failure modes break the continuity of R_t:

1. **Subspace Ambiguity (Eigengap Collapse):** At points where the spectral gap vanishes ($\lambda_d - \lambda_{d+1} \to 0$), the normal projector P_N becomes ill-defined. Consequently, the normal gradient condition $\|P_N \nabla \log p_t\| = 0$ becomes ill-posed as the normal eigenframe loses its orientation.

2. **Topological Discontinuities:** At locations where $\lambda_{d+1}(y) = 0$, the concavity condition triggers abruptly, causing R_t to instantly gain or lose entire connected components.



(a) Ridge of an asymmetric circle density at $\sigma = 0.54$

(b) Ridge of an asymmetric circle density at $\sigma = 0.58$

Figure 2: Comparison of ridge structures for an asymmetric semicircle density mixture at different noise levels σ. A small change in σ alters the topological structure of the ridge.

Genovese et al. (2014) address these failure modes by proving stability bounds under an assumed lower bound $\beta > 0$ on the spectral gap. Nevertheless, the underlying instability is intrinsic to the map $(m_t, A_t) \mapsto R_t$: passing smooth analytic fields through rigid scalar thresholding functions inevitably induces spatial and temporal discontinuities.

A second structural limitation is that the ridge formulation requires fixing the intrinsic dimension $d \in \mathbb{N}$ a priori. In practice, there is no single consistent choice of d across all noise scales. While dimension is a required integer parameter for normal frame projection, the natural scale-dependent effective dimension:

$$d_{\text{eff}}(t) = \mathbb{E}\left[\operatorname{tr} A_t(X_t)\right] \tag{56}$$

is neither integer-valued nor constant in t. Instead, $d_{\mathrm{eff}}(t)$ continuously interpolates between the intrinsic dimension d at small noise scales and zero at large noise scales, often overshooting non-monotonically at intermediate scales (section 5).

Example: Discontinuity of the Ridge Set. Consider a probability distribution supported on the unit circle $\mathbb{S}^1 \subset \mathbb{R}^2$, formed by a $1 : 3$ mixture of uniform densities along the left and right semicircles, respectively. As shown in Figure 2, a minute increase in the smoothing parameter σ causes a drastic shift in both the spatial position and topology of the ridge. Furthermore, knowing the geometry of R_σ provides no information regarding how mass is distributed along it; the probability density may be negligible over large portions of the ridge set.

Conclusion on the Ridge. In summary, while the ridge set R_σ provides a useful theoretical surrogate for $\mathcal{M}$ at small noise levels σ and serves as the local attractor of the score field, it is ill-behaved, less helpful, and discontinuous at larger noise scales. Recognising that the ridge is fundamentally a functional of the Jacobian spectrum of the score leads naturally to the continuous formulation presented in section 5.

5 Posterior Covariance Field

Here we introduce the ultimate object we analyse: posterior covariance fields.

Definition 5.1. Let $Y_t = X + \sqrt{t}Z$, $m_t(y) := \mathbb{E}\left[X \mid Y_t = y\right]$, and

$$A_t(y) := \frac{1}{t}\operatorname{Cov}\left(X \mid Y_t = y\right).$$

We call the weighted object $\mathfrak{F}_t := \operatorname{Law}\left(Y_t, m_t\left(Y_t\right), A_t\left(Y_t\right)\right)$ the *posterior covariance field*.

There are a few reasons why it makes sense to look at this object. First, it is not plagued by the issues we encountered in the analysis of ridges or level sets in section 4. These objects suffered from being too geometric for our use case, where we care about noise scales much greater than 0. We need a more analytic quantity, and the posterior covariance field is defined for all noise scales $t > 0$. In fact level sets and ridges can just be viewed as functionals of $\mathfrak{F}_t$.

Second, the posterior covariance captures the local geometry of the score. The posterior covariance corresponds to the Jacobian of the score. This link is given in section 2. This fact, which is not at all obvious, is where we reap the rewards. Here we normalise the posterior covariance by t to account for the scale of noise added.

Third, here we deal with expectations, rather than L^∞-type quantities like the reach τ_M. If there exists a region of the density which is geometrically complex and as a consequence has poor regularity, but which carries almost no mass, we should not let it dramatically change our understanding of the behaviour of the score.

5.1 Three functionals of the spectrum

All three quantities below are moments of the same object: the spectral measure of A_t, averaged over $y \sim p_t$. Write $a_1(y), \ldots, a_D(y)$ for the eigenvalues of $A_t(y)$ and let $\mu_t := \operatorname{Law}\big(a_I(Y_t)\big)$ with I uniform on $\{1, \ldots, D\}$, so that $\mathbb{E}\sum_i f(a_i) = D\int f\,d\mu_t$ for any f.

5.1.1 Effective dimension

Definition 5.2.

$$d_{\text{eff}}(t) \ := \ \mathbb{E}\big[\operatorname{tr} A_t(Y_t)\big] \ = \ \mathbb{E}\sum_i a_i. \tag{57}$$

Proposition 5.3 (Exact MMSE identity)**.** *Let* $\operatorname{MMSE}(t) := \mathbb{E}\big\|X - m_t(Y_t)\|^2$. *Then*

$$d_{\text{eff}}(t) \ = \ \frac{\operatorname{MMSE}(t)}{t}. \tag{58}$$

Proof. $\operatorname{tr}\operatorname{Cov}(X \mid Y_t) = \mathbb{E}\big[\|X - m_t(Y_t)\|^2 \mid Y_t\big]$; take expectations and divide by t. □

Note in identity (58) MMSE(t) is precisely the population value of the ε-prediction (or x_0-prediction) denoising loss at noise level t, so it requires no computation beyond the training objective already being evaluated.

Link to dimension. An orthogonal projector of rank k has spectrum $\{1^{(k)}, 0^{(D-k)}\}$ and trace k. In the ideal manifold limit $A_t \approx P_{T_x\mathcal{M}}$ and hence $d_{\text{eff}}(t) \approx d$. Equivalently, via (58): a d-dimensional manifold leaves $O(t)$ posterior uncertainty in each of d tangent directions and $o(t)$ in the normal directions, so MMSE $\approx d \times t$. Each eigenvalue may be read as a fractional dimension: $a_i = 0.4$ contributes 0.4 effective directions.

Failure mode. d_{eff} is a first moment and therefore cannot distinguish $\text{diag}(1,1,1,0,0,0)$ from $\text{diag}(\frac{1}{2},\frac{1}{2},\frac{1}{2},\frac{1}{2},\frac{1}{2},\frac{1}{2})$: both have trace 3. It also overcounts when $a_i > 1$, in which case the "fractional dimension" reading breaks down entirely. The next two functionals repair these two deficiencies.

5.1.2 Distance to projector

Definition 5.4.

$$\Delta_{\text{proj}}(t) \;:=\; \mathbb{E}\sum_i \min\bigl\{a_i^2,\ (a_i-1)^2\bigr\}. \tag{59}$$

The summand is the squared distance from a_i to the set $\{0,1\}$, and the two branches exchange at $a_i = \frac{1}{2}$:

$$\min\{a^2,(a-1)^2\} = \begin{cases} a^2, & a \le \frac{1}{2} \quad \text{(read as normal-like)},\\ (a-1)^2, & a \ge \frac{1}{2} \quad \text{(read as tangent-like)}. \end{cases} \tag{60}$$

Proposition 5.5. *Let $\mathcal{P} = \{P \in \mathbb{R}^{D\times D} : P = P^\top = P^2\}$ be the set of orthogonal projectors and let A be symmetric with eigenvalues a_i. Then*

$$\min_{P\in\mathcal{P}} \|A-P\|_F^2 \;=\; \sum_i \min\{a_i^2,(a_i-1)^2\}, \qquad \textit{hence} \qquad \Delta_{\text{proj}}(t) = \mathbb{E}\bigl[\text{dist}_F^2\bigl(A_t(Y_t),\mathcal{P}\bigr)\bigr]. \tag{61}$$

Proof. $\mathcal{P}$ is invariant under conjugation by orthogonal matrices, and for symmetric A the Frobenius distance to a conjugation-invariant set is minimised by a P sharing the eigenvectors of A (a consequence of the Hoffman–Wielandt inequality). It then remains to round each a_i to the nearer of 0 or 1. □

Thus Δ_{proj} measures, on average, how far the soft posterior geometry is from being a genuine tangent-space projector. It vanishes exactly when $A_t(Y_t)$ is a projector almost surely.

5.1.3 Ambiguity

Definition 5.6. With $(x)_+ := \max\{x, 0\}$,

$$\mathcal{A}(t) \;:=\; \mathbb{E} \sum_i \bigl(a_i - 1\bigr)_+^2. \tag{62}$$

The motivation is dynamical rather than geometric. Let (a_i, v_i) be an eigenpair of A_t. $a_i > 1$ occurs if and only if the reverse probability-flow ODE is locally expansive along v_i. The threshold where posterior variance exceeds the injected noise variance:

$$a_i = \frac{\text{posterior variance along } v_i}{\text{noise variance } t} \qquad \begin{cases} < 1: & \text{denoising, contraction,} \\ = 1: & \text{neutral,} \\ > 1: & \text{amplification, expansion.} \end{cases} \tag{63}$$

The three design choices are then: threshold at 1 (only expansion is penalised), measure the *excess* $a_i - 1$ (the neutral point is 1, not 0), and square it (making the functional comparable to a squared Frobenius norm and penalising strong expansion super-linearly).

Link to ambiguity. Suppose the posterior at y is approximately $\frac{1}{2}\mathcal{N}(\mu_1, \Sigma) + \frac{1}{2}\mathcal{N}(\mu_2, \Sigma)$, corresponding to two competing clean configurations. Then

$$\operatorname{Cov}(X \mid Y_t = y) \;=\; \Sigma + \tfrac{1}{4}(\mu_1 - \mu_2)(\mu_1 - \mu_2)^\top, \tag{64}$$

and the between-mode term can exceed tI along $\mu_1 - \mu_2$, giving $a_i > 1$. So $a_i > 1$ signals that the noisy observation fails to identify a single locally stable latent configuration.

Remark 5.7 (Scope of interpretation). Strictly, $a_i > 1$ is a diagnostic that posterior variance exceeds the noise scale. Multimodality is the typical, but not the only, mechanism: heavy-tailed unimodal p_0 (e.g. Laplace, Student-t) also produces $a_i > 1$. "Ambiguity" is the geometric reading of the phenomenon, not a theorem.

Remark 5.8 (Mean versus tail). $\mathcal{A}$ is an average over $y \sim p_t$ and can be small while $\sup_y a_{\max}(y)$ is large, if the expansive region carries little mass. In practice one should report both $\mathcal{A}(t)$ and a high quantile of $(a_{\max} - 1)_+$, since sampler failure is driven by trajectories that enter the expansive region, not by its measure.

5.2 Evolution law of MMSE

By Theorem 6.1, differentiating $\operatorname{MMSE}(t)$ under the Gaussian channel model $Y_t = X + \sqrt{t}Z$ gives:

$$\operatorname{MMSE}'(t) = \frac{1}{t^2}\mathbb{E}\left[\operatorname{tr}\left(\operatorname{Cov}\left(X \mid Y_t\right)^2\right)\right] \tag{65}$$

By definition, $A_t(y) = \frac{1}{t}\operatorname{Cov}(X \mid Y_t = y)$. Squaring this relation and taking the trace yields:

$$\operatorname{tr}\left(A_t(Y_t)^2\right) = \frac{1}{t^2}\operatorname{tr}\left(\operatorname{Cov}(X \mid Y_t)^2\right) = \|A_t(Y_t)\|_F^2 \tag{66}$$

Substituting this directly into the expression for $\mathrm{MMSE}'(t)$:

$$\mathrm{MMSE}'(t) = \mathbb{E}\left[\operatorname{tr}\left(A_t(Y_t)^2\right)\right] = \mathbb{E}\|A_t(Y_t)\|_F^2 \tag{67}$$

5.3 MMSE-driven discretisation

We consider the additive Gaussian perturbation

$$X_t = X_0 + \sqrt{t}\,Z, \qquad Z \sim \mathcal{N}(0, I_D), \tag{68}$$

so that, in the notation of the diffusion model,

$$c_t = 1, \qquad \sigma_t^2 = t. \tag{69}$$

Let

$$m_t(x) := \mathbb{E}[X_0 \mid X_t = x] \tag{70}$$

denote the posterior mean and define the minimum mean-squared error

$$\mathrm{MMSE}(t) := \mathbb{E}\left[\|X_0 - m_t(X_t)\|^2\right]. \tag{71}$$

Following the effective-dimension viewpoint, we write

$$d_{\mathrm{eff}}(t) := \frac{\mathrm{MMSE}(t)}{t}, \tag{72}$$

and hence

$$\mathrm{MMSE}(t) = t\, d_{\mathrm{eff}}(t). \tag{73}$$

Discretisation error. For additive Gaussian noise, Tweedie's formula gives

$$s_t(x) = \frac{m_t(x) - x}{t}. \tag{74}$$

The same martingale argument used in (Potaptchik et al., 2024) to control the score variation in the discretised reverse process then yields, for two noise levels $t < t'$,

$$\begin{aligned} &\mathbb{E}\left\|s_t(X_t \mid t', X_{t'}) - s_t(X_t)\right\|^2 \\ &\quad = \frac{1}{t^2}\left(\mathrm{MMSE}(t') - \mathrm{MMSE}(t)\right), \end{aligned} \tag{75}$$

with the sign understood according to the direction in which the diffusion trajectory is traversed. Thus, for a sufficiently fine discretisation with step size

$$\Delta t_k := t_{k+1} - t_k, \tag{76}$$

the contribution of the k-th interval is, up to higher-order terms,

$$\mathcal{E}_k \asymp \frac{\left|\mathrm{MMSE}'(t_k)\right|}{t_k^2}(\Delta t_k)^2. \tag{77}$$

Consequently, the total discretisation error is approximated by

$$\mathcal{E}_{\mathrm{disc}} \asymp \sum_{k=0}^{K-1} \alpha(t_k)(\Delta t_k)^2, \qquad \alpha(t) := \frac{|\,\mathrm{MMSE}'(t)|}{t^2}. \tag{78}$$

Optimal allocation of time steps. We now choose the discretisation points to minimise (78) subject to a fixed number K of steps. Introduce the local step density

$$n(t) := \frac{1}{\Delta t(t)}. \tag{79}$$

In the continuum limit,

$$K \approx \int n(t)\,dt, \tag{80}$$

while

$$\begin{aligned} \mathcal{E}_{\mathrm{disc}} &\approx \int \alpha(t)\Delta t(t)\,dt \\ &= \int \frac{\alpha(t)}{n(t)}\,dt. \end{aligned} \tag{81}$$

Minimising this functional subject to $\int n(t)\,dt = K$ gives

$$n(t) \propto \sqrt{\alpha(t)}, \tag{82}$$

and therefore

$$\Delta t(t) \propto \frac{1}{\sqrt{\alpha(t)}} = \frac{t}{\sqrt{|\,\mathrm{MMSE}'(t)|}}. \tag{83}$$

Finally, using Equation 67,

$$\mathrm{MMSE}'(t) = \mathbb{E}\left\|A_t\left(Y_t\right)\right\|_F^2, \tag{84}$$

so the resulting MMSE-adaptive discretisation is

$$\Delta t(t) \propto \frac{t}{\sqrt{\mathbb{E}\left\|A_t\left(Y_t\right)\right\|_F^2}}. \tag{85}$$

5.4 Examples and Interpretation

In Figure 3 we plot $d_{\mathrm{eff}}, \Delta_{\mathrm{proj}}, \mathcal{A}$, along with $\frac{t}{\sqrt{\mathbb{E}\|A_t(Y_t)\|_F^2}}$ for three toy densities: Bernoulli($\{+1,-1\}$), Uniform($\mathcal{S}^1$), and Uniform($\{\mathcal{S}^1\} \times \{\frac{1}{4}, -\frac{1}{4}\}$). As expected, $d_{\mathrm{eff}}, \Delta_{\mathrm{proj}}, \mathcal{A}$ vary over time and capture different behaviours. We also see this has an impact on the natural time-step size allocation given by $\frac{t}{\sqrt{\mathbb{E}\|A_t(Y_t)\|_F^2}}$.

In the Bernoulli($\{+1,-1\}$) case we see d_{eff} climb towards 1 as the noised density rises in $[-1,1]$, until the scale of the noise added dominates and d_{eff} returns to 0. (Note: in general for small time d_{eff} is the true dimension, and for large time $d_{\text{eff}}=0$.) Similarly, for small time we are in a projector-like regime. The first bump in projector distance is due to $\mathcal{A}$, and the second is due to the scale of noise added. The behaviour is largely similar in the Uniform($\mathcal{S}^1$) setting.

The change in Uniform($\{\mathcal{S}^1\}\times\{\frac{1}{4},-\frac{1}{4}\}$) can be explained by the two critical times in the diffusion process: first, when the two circles merge; and second, when they collapse. The merging of the two circles makes our density cylinder-like, causing d_{eff} to exceed 1 and tend to 2 until the noise scale becomes too large.

The time-step density plots interestingly capture the discrete nature of the Bernoulli($\{+1,-1\}$) case. We see that many more small timesteps are required for larger t, as this is where a "phase transition" occurs and points pick a final destination: +1 or -1. However, this behaviour is not observed in the Uniform($\mathcal{S}^1$) and Uniform($\{\mathcal{S}^1\}\times\{\frac{1}{4},-\frac{1}{4}\}$) cases as the support is no longer discrete. We do see that dimension plays a role in the slope of the curve, which is expected since for a manifold $\|A_t(Y_t)\|_F^2=\sum_i a_i^2=\sum_i \mathbb{1}\{a_i=1\}=\dim\mathcal{M}$.

Remark 5.9. While here $d_{\text{eff}},\Delta_{\text{proj}},\mathcal{A}$ are used, note that they are expectations, and as such their distribution is more informative in some cases. For example, if the support is the disjoint union of $\mathbb{S}^1$ and $\mathbb{S}^2$, each with equal weight, then $d_{\text{eff}}=\frac{1}{2}(1+2)=\frac{3}{2}$. The density plot of $\sum_i a_i$ will show modes at the different dimensions of the support. Similar subtleties can be uncovered for $\Delta_{\text{proj}},\mathcal{A}$.

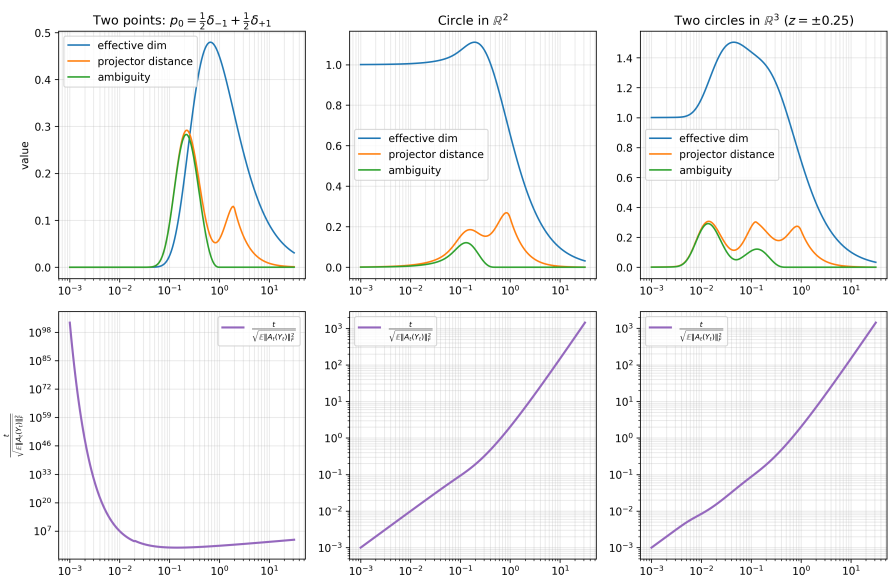


Figure 3: Functionals and assigned time step evolution for Bernoulli($\{+1,-1\}$), Uniform($\mathcal{S}^1$), and Uniform($\{\mathcal{S}^1\}\times\{\frac{1}{4},-\frac{1}{4}\}$).

Geometric Interpretation. We see the effect of the $\mathrm{MMSE}'(t)$ term on time-step density assignment. Notably, since $t d_{\mathrm{eff}}(t) = \mathrm{MMSE}(t)$, we have $\mathrm{MMSE}'(t) = d_{\mathrm{eff}}(t) + t d'_{\mathrm{eff}}(t)$. Hence, higher d_{eff}, and especially changes in d_{eff} at larger t, require smaller time steps. Furthermore, $\mathbb{E}\left\|A_t\left(Y_t\right)\right\|_F^2 = \sum_i \mathbb{E}\left[a_i(t)^2\right]$, so this discretisation assigns a high density of small time-steps whenever we have expansive eigenvalues $a_i > 1$, in expectation.

Dynamics Interpretation. The perhaps clearest interpretation is from the viewpoint of dynamics: when the score field does not change much in time, we can take a larger step in time without incurring a large error. When the score varies drastically in time, this is exactly when we need a higher density of time steps to ensure that we do not incur a high discretisation error.

Probabilistic Interpretation. Time regularity of the score $\nabla \log p_t$ translates into the time regularity of MMSE, via the posterior mean m_t. Hence it is precisely when our posterior mean and MMSE are changing that the score changes substantially, therefore requiring a higher density of time steps.

Remark 5.10. There also is an information-theoretic interpretation. By (Guo et al., 2005) we have

$$\frac{d}{dt} I\left(X_0; X_t\right) = \frac{dI}{d\gamma} \cdot \frac{d\gamma}{dt} = \left(\frac{1}{2}\,\mathrm{MMSE}(t)\right)\left(-\frac{1}{t^2}\right) = -\frac{\mathrm{MMSE}(t)}{2t^2}$$

where γ is the signal-to-noise ratio (here $1/t$). We do not yet fully understand this, and are continuing to investigate this. We expect the conclusion to relate time step density to information flow.

Conclusion on the Posterior Covariance Field. The spectrum of the posterior covariance (and its distribution) is smooth and highly informative. Many possible metrics can be built from it, including a measure of dimension d_{eff}. The spectrum of the posterior covariance relates closely to the Jacobian of the score, as discussed in subsection 2.3, and as such determines the dynamical stability of the denoising process. The expected Frobenius norm of the rescaled posterior covariance controls step size when discretising and solving the velocity field ODE in generation, and yields an optimal choice of time schedule.

6 Conclusion and Next Steps

Throughout this research internship we have attempted to build the tools necessary to understand how the geometry of a density changes as it is noised. This is key in understanding diffusion-based generative models. We explored anisotropic error propagation, level sets and ridges, and the spectrum of the posterior covariance.

After these successive attempts to find an informative and tractable object, we conclude that the spectrum of the posterior covariance, and functions thereof, are key objects. This allows us to build metrics to analyse the behaviour of densities throughout the noising process, and sheds new light on where complexity lies during the denoising process.

The commonly held belief that score regularity decreases as we approach our final density p_{data} is incomplete, and sometimes incorrect. Changes in geometry as a result of noise affect score regularity, and this can happen at intermediate (non-final) times during generation.

Next steps. We would like to investigate the behaviour of $d_{\text{eff}}, \Delta_{\text{proj}}, \mathcal{A}$ for real datasets and trained score networks. Furthermore, we would like to investigate latent diffusion models to see how $d_{\text{eff}}, \Delta_{\text{proj}}, \mathcal{A}$ vary between lower-dimensional latent space and higher dimensional input/output space. We also would like to explore what else can be gained from the spectrum of the poserior covariance.

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AI Usage

Image to text converters were used to aid in typesetting equations, and large language models were used for some code generation and in editing parts of the report.

References

Eddie Aamari and Clément Berenfeld. A theory of stratification learning. *arXiv preprint arXiv:2405.20066*, 2024.

Michael Albergo, Nicholas M Boffi, and Eric Vanden-Eijnden. Stochastic interpolants: A unifying framework for flows and diffusions. *Journal of Machine Learning Research*, 26(209):1–80, 2025.

Joe Benton, Valentin De Bortoli, Arnaud Doucet, and George Deligiannidis. Nearly d-linear convergence bounds for diffusion models via stochastic localization. In *International Conference on Learning Representations*, volume 2024, pages 36916–36936, 2024.

Bradley CA Brown, Anthony L Caterini, Brendan Leigh Ross, Jesse C Cresswell, and Gabriel Loaiza-Ganem. The union of manifolds hypothesis. In *NeurIPS 2022 Workshop on Symmetry and Geometry in Neural Representations*, 2022a.

Bradley CA Brown, Anthony L Caterini, Brendan Leigh Ross, Jesse C Cresswell, and Gabriel Loaiza-Ganem. Verifying the union of manifolds hypothesis for image data. *arXiv preprint arXiv:2207.02862*, 2022b.

Herbert Federer. Curvature measures. *Transactions of the American mathematical Society*, 93(3):418–491, 1959.

Christopher R Genovese, Marco Perone-Pacifico, Isabella Verdinelli, and Larry Wasserman. Nonparametric ridge estimation. 2014.

Dongning Guo, Shlomo Shamai, and Sergio Verdú. Mutual information and minimum mean-square error in gaussian channels. *IEEE transactions on information theory*, 51(4):1261–1282, 2005.

Jonathan Ho, Ajay Jain, and Pieter Abbeel. Denoising diffusion probabilistic models. *Advances in neural information processing systems*, 33:6840–6851, 2020.

Jiadong Liang, Zhihan Huang, and Yuxin Chen. Low-dimensional adaptation of diffusion models: Convergence in total variation. *arXiv preprint arXiv:2501.12982*, 2025.

Yaron Lipman, Ricky TQ Chen, Heli Ben-Hamu, Maximilian Nickel, and Matt Le. Flow matching for generative modeling. *arXiv preprint arXiv:2210.02747*, 2022.

Peter Potaptchik, Iskander Azangulov, and George Deligiannidis. Linear convergence of diffusion models under the manifold hypothesis. *arXiv preprint arXiv:2410.09046*, 2024.

Jiaming Song, Chenlin Meng, and Stefano Ermon. Denoising diffusion implicit models. *arXiv preprint arXiv:2010.02502*, 2020a.

Yang Song and Stefano Ermon. Generative modeling by estimating gradients of the data distribution. *Advances in neural information processing systems*, 32, 2019.

Yang Song, Jascha Sohl-Dickstein, Diederik P Kingma, Abhishek Kumar, Stefano Ermon, and Ben Poole. Score-based generative modeling through stochastic differential equations. *arXiv preprint arXiv:2011.13456*, 2020b.

Appendix: MMSE Evolution

Theorem 6.1. *Let $X \in \mathbb{R}^D$ be a random vector with probability density $p(x)$ and finite second moment, and let $Z \sim \mathcal{N}(0, I_D)$ be independent of X. Define the observation process $Y_t = X + \sqrt{t}Z$ for $t > 0$, and the Minimum Mean Squared Error function:*

$$\mathrm{MMSE}(t) = \mathbb{E}\left[\|X - \mathbb{E}[X \mid Y_t]\|^2\right]$$

Then $\mathrm{MMSE}(t)$ *is differentiable with respect to $t > 0$, and its derivative satisfies:*

$$\mathrm{MMSE}'(t) = \frac{1}{t^2}\mathbb{E}\left[\mathrm{tr}\left(\mathrm{Cov}(X \mid Y_t)^2\right)\right]$$

Proof. The marginal density of Y_t, $p_t(y) = \int p(x)(2\pi t)^{-D/2} \exp\left(-\frac{\|y-x\|^2}{2t}\right) dx$, satisfies the heat equation $\frac{\partial p_t(y)}{\partial t} = \frac{1}{2}\Delta_y p_t(y)$. Define the score function $s_t(y) = \nabla_y \log p_t(y)$ and the Fisher information trace $J(t) = \mathbb{E}\left[\|s_t(Y_t)\|^2\right]$.

1. Score representation of conditional statistics. Spatial differentiation of $p_t(y)$ yields Tweedie's formula:

$$s_t(y) = \frac{\nabla_y p_t(y)}{p_t(y)} = -\frac{y - \mathbb{E}[X \mid Y_t = y]}{t} \implies X - \mathbb{E}[X \mid Y_t] = -ts_t(Y_t)$$

Taking the spatial Jacobian of $\mathbb{E}[X \mid Y_t = y] = y + ts_t(y)$:

$$\nabla_y \mathbb{E}[X \mid Y_t = y] = \frac{1}{t}\,\mathrm{Cov}(X \mid Y_t = y) = I_D + t\nabla_y s_t(y)$$

Rearranging yields the score Hessian:

$$\nabla_y s_t(y) = \frac{\mathrm{Cov}(X \mid Y_t = y) - tI_D}{t^2}$$

2. Relation between MMSE and Fisher information. Since $\mathbb{E}[\|s_t(Y_t)\|^2] = -\mathbb{E}[\mathrm{tr}\,\nabla_y s_t(Y_t)]$ and $\mathbb{E}[\mathrm{tr}\,\mathrm{Cov}(X \mid Y_t)] = \mathrm{MMSE}(t)$, substituting the score Hessian gives

$$J(t) = -\mathbb{E}\left[\mathrm{tr}\,\frac{\mathrm{Cov}(X \mid Y_t) - tI_D}{t^2}\right] = \frac{D}{t} - \frac{\mathrm{MMSE}(t)}{t^2} \implies \mathrm{MMSE}(t) = tD - t^2 J(t).$$

3. Time evolution of Fisher information. Differentiating $J(t) = \int p_t(y)\|\nabla \log p_t(y)\|^2 dy$ with respect to t under heat flow $\frac{\partial p_t}{\partial t} = \frac{1}{2}\Delta p_t$ yields de Bruijn's score identity:

$$J'(t) = -\mathbb{E}\left[\|\nabla_y s_t(Y_t)\|_F^2\right] = -\mathbb{E}\left[\mathrm{tr}\left((\nabla_y s_t(Y_t))^2\right)\right]$$

Substituting the expression for $\nabla_y s_t(Y_t)$:

$$\begin{aligned}
J'(t) &= -\mathbb{E}\left[\operatorname{tr}\left(\left(\frac{\operatorname{Cov}(X \mid Y_t) - tI_D}{t^2}\right)^2\right)\right] \\
&= -\frac{1}{t^4}\mathbb{E}\left[\operatorname{tr}\left(\operatorname{Cov}(X \mid Y_t)^2 - 2t\operatorname{Cov}(X \mid Y_t) + t^2 I_D\right)\right] \\
&= -\frac{1}{t^4}\mathbb{E}\left[\operatorname{tr}\left(\operatorname{Cov}(X \mid Y_t)^2\right)\right] + \frac{2}{t^3}\operatorname{MMSE}(t) - \frac{D}{t^2}
\end{aligned}$$

4. Total differentiation of MMSE(t). Differentiating $\operatorname{MMSE}(t) = tD - t^2 J(t)$ via the product rule:

$$\operatorname{MMSE}'(t) = D - 2tJ(t) - t^2 J'(t)$$

Substituting $2tJ(t) = 2D - \frac{2\operatorname{MMSE}(t)}{t}$ and $-t^2 J'(t)$:

$$\begin{aligned}
\operatorname{MMSE}'(t) &= D - \left(2D - \frac{2\operatorname{MMSE}(t)}{t}\right) + \left(\frac{1}{t^2}\mathbb{E}\left[\operatorname{tr}\left(\operatorname{Cov}(X \mid Y_t)^2\right)\right] - \frac{2\operatorname{MMSE}(t)}{t} + D\right) \\
&= (D - 2D + D) + \left(\frac{2\operatorname{MMSE}(t)}{t} - \frac{2\operatorname{MMSE}(t)}{t}\right) + \frac{1}{t^2}\mathbb{E}\left[\operatorname{tr}\left(\operatorname{Cov}(X \mid Y_t)^2\right)\right] \\
&= \frac{1}{t^2}\mathbb{E}\left[\operatorname{tr}\left(\operatorname{Cov}(X \mid Y_t)^2\right)\right]
\end{aligned}$$

□